

Green's function analysis of tunnel-coupled topological interface states

Abstract

This work applies an analytical Green's function framework for interface states between two semi-infinite systems, following the formalism of Garcia-Moliner and Velasco, and studies its application across arbitrary tunnel-coupling regimes. The framework provides a transparent, efficient tool for analysing coupled quantum systems.

1 Introductions

Surface states belong to the most distinctive signatures of topological phases of matter [1]. Representative examples include the gapless surface states of topological insulators [2] and the Majorana bound states in topological superconductors [3], whose existence is guaranteed by the corresponding topological invariants. In most theoretical treatments, such states are considered in isolation, typically at the boundary between a topological material and vacuum. In this setting, bulk-boundary correspondence [4] provides a robust framework that allows one to predict the existence and number of boundary states directly from bulk topological invariants. This correspondence has been extensively studied across a wide range of models.

In realistic quantum devices and experimental settings, we often encounter interfaces between distinct materials rather than boundaries with vacuum. When two materials are brought into contact, the resulting interface problem becomes significantly more complicated. A variety of methods have been developed to study such systems. Numerical approaches, such as iterative schemes [5], can accurately resolve the interface spectrum, but typically provide limited analytical insight. For three-dimensional quantum materials, slab model offer an exact numerical treatment [6], but suffer from interactions between the two surface and finite-size convergence issues. Perturbative approaches can capture weak hybridisation effects [7], but break down once the coupling strength becomes comparable to the bulk energy scales.

An analytical Green's function formalism for material-material interfaces was developed by F. Garcia-Moliner and V. R. Velasco [8] in 1986; in this study, we reformulate this method in a notation suited to tight-binding systems. Under this formulation, the contribution of each subsystem to the interface problem is encoded in its bulk Green's function, allowing the interface Green's function to be constructed without diagonalising the complete coupled system. This thereby significantly reduces the complexity of the problem while retaining its essential physics.

To demonstrate the utility of this formalism, we apply it to representative tight-binding models and study the evolution of interface states from weak to strong tunnel coupling. We first consider the SSH chain as the simplest one-dimensional topological system, and systematically track the generation, hybridisation, redistribution and disappearance of interface states as the tunnel coupling strength varies. In the strong-tunnel-coupling limit, a simple picture emerges in terms of dimerisation and effective boundary reconstruction. We then extend the analysis to a two-dimensional chiral superconductor. By examining different topological configurations with different chirality, we uncover special behaviours of the interface, including gap opening and mode scattering.

Section 2 reviews and reformulates the established interface Green's function formalism. Section 3 and 4 contain the main applications of this project, where the method is implemented for coupled SSH chains and $p + ip$ superconductors and used to analyse the resulting interface spectra.

2 Derivation of interface Green's functions

In this section, we rederive and reformulate the interface Green's function formalism of Garcia-Moliner and Velasco[8] for two semi-infinite systems coupled at their interface, using notation suited to the tight-binding applications considered below.

Consider two semi-infinite systems connected via a local tunnel coupling at the interface, as shown in Figure 1. The Hilbert space naturally decomposes into a direct sum of left and right subspaces, $\mathcal{H} = \mathcal{H}^L \oplus \mathcal{H}^R$. We introduce projection operator P^L and P^R onto the corresponding

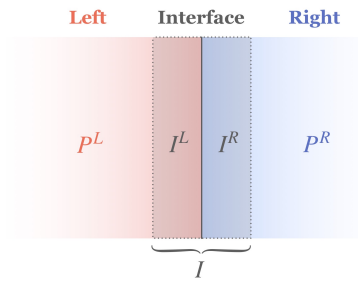


Figure 1: Two semi-infinite systems joined at an interface, with the subspace projectors indicated. $P^{L,R}$: projection onto the left/right semi-infinite part. $I^{L,R}$: projection onto the left/right interface. I : projection onto the complete interface.

subspaces, satisfying $P^L + P^R = \mathbb{1}$, see Figure 1. In this representation, the Hamiltonian can be written in block form¹ as

$$H = \begin{pmatrix} P^L h^L P^L & P^L V^\dagger P^R \\ P^R V P^L & P^R h^R P^R \end{pmatrix}, \quad (1)$$

where $h^{L,R}$ denote the bulk Hamiltonian of the left and right semi-infinite systems respectively, and V describes the local tunnel coupling across the interface. Our goal is to construct the Green's function [9] of the coupled system,²

$$G(\omega) = (\omega \mathbb{1} - H)^{-1}, \quad (2)$$

and especially concern its projection onto the interface subspace [8], since all physical information about the interface states is encoded in this sector.

In order to extract the interfacial physics, we introduce projection operator I^L and I^R acting on the interface subspaces of the left and right system, with $I^L + I^R = I$, see Figure 1. Direct inversion in the full semi-infinite space may be ill-conditioned, so we rewrite Eq. (2) into $(\omega \mathbb{1} - H)G = \mathbb{1}$ and projecting it onto the interface subspace, yields

$$\omega IGI - IHGI = I. \quad (3)$$

¹See Appendix A.2.

²Note that we work in units where $\hbar = 1$ throughout this work. Factors of $\hbar$ can be restored by dimensional analysis if required.

We define the interface Green's function as $G_I \equiv IGI$. If the term $IHGI$ can be expressed solely in terms of G_I , one obtains a closed effective matrix equation of the form $MG_I = I$, where M is an to-be-determined matrix from which the interface Green's function can be explicitly solved.

As a good starting point, we define the Green's function for the uncoupled system as

$$g(\omega) = \begin{pmatrix} P^L g^L P^L & 0 \\ 0 & P^R g^R P^R \end{pmatrix}, \quad (4)$$

where $g^{L,R}(\omega) = (\omega \mathbb{1} - h^{L,R})^{-1}$ is the bulk Green's function of left and right subspaces respectively. The coupling between the two subsystems is incorporated via Dyson equation,

$$G = g + g\Sigma g, \quad (5)$$

where Σ denotes the self-energy arising from interfacial scattering. Restricted to the interface subspace, the self-energy can be written as

$$\Sigma = I\Sigma I = \begin{pmatrix} I^L L I^L & I^L T^\dagger I^R \\ I^R T I^L & I^R R I^R \end{pmatrix}, \quad (6)$$

where R, L describe reflection processes at the interface for the left and right subsystems respectively, and T denotes the transmission from left to right, see Figure 2.

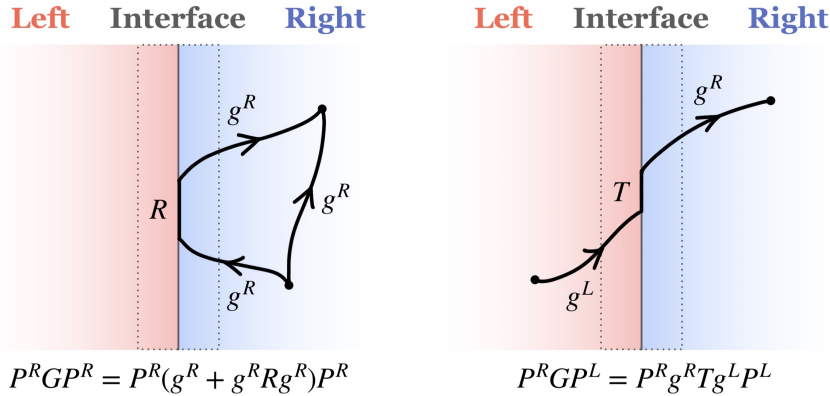


Figure 2: Schematic illustration of the effect of a material-material interface on the trajectory of Green's function. Left: reflection at the interface. Right: transmission across the surface. These contributions are naturally organised using the Eq. (5).

To eliminate the explicit dependence on Σ and obtain a closed equation, we project Eq. (5) onto the interface subspace:

$$IGI = IgI + IgI\Sigma IgI. \quad (7)$$

Rearranging this expression yields

$$\Sigma = (IgI)^{-1} I(G - g) I (IgI)^{-1}, \quad (8)$$

assuming that $g_I \equiv IgI$ is invertible³ within the corresponding subspace. Substituting this back into Eq. (5) gives

$$G = g + g(IgI)^{-1} I(G - g) I (IgI)^{-1} g. \quad (9)$$

³For a rigorous discussion of the matrix inverse in the context of coupled semi-infinite system, see Appendix A.3.

Projecting onto the interface from the right side, we obtain

$$\begin{aligned}
GI &= gI + g(IGI)^{-1}I(G - g)I(IGI)^{-1}gI \\
&= gI + g(IGI)^{-1}IGI - g(IGI)^{-1}IgI \\
&= g(IGI)^{-1}G_I,
\end{aligned} \tag{10}$$

since we have $(IGI)^{-1}gI = I$ as a corollary⁴, and the invertibility of IGI has been used. Similarly, one finds $IG = G_I(IGI)^{-1}g$ for completeness.

Substituting Eq. (10) into Eq. (3), we obtain

$$\omega G_I - IHg(IGI)^{-1}G_I = I. \tag{11}$$

It follows that the interface Green's function satisfies

$$(G_I)^{-1} = \omega I - IHg(IGI)^{-1}. \tag{12}$$

Eq. (12) reproduces the interface Green's function obtained by Garcia-Moliner and Velasco, here expressed in the projector notation used throughout this work. More generally, this can be written in block form explicitly as

$$\begin{pmatrix} G_I^{LL} & G_I^{LR} \\ G_I^{RL} & G_I^{RR} \end{pmatrix}^{-1} = \begin{pmatrix} \omega I^L - I^L H^{LL} g^L (I^L g^L I^L)^{-1} & -I^L H^{LR} I^R \\ -I^R H^{RL} I^L & \omega I^R - I^R H^{RR} g^R (I^R g^R I^R)^{-1} \end{pmatrix}. \tag{13}$$

The simplification of the off-diagonal blocks relies on the principal layer assumption, namely that the interface tunnel coupling is localised entirely at the boundary degrees of freedom,

$$H^{M\bar{M}} = I^M H^{M\bar{M}} I^{\bar{M}}, \tag{14}$$

where $M \in \{L, R\}$ and $\bar{M}$ denotes the opposite subsystem.

In the limiting case where one side is vacuum (e.g., the left subsystem is absent, such that $H^{LL} = H^{LR} = H^{RL} = 0$), Eq. (13) reduces to the surface Green's function of an isolated system,

$$(G_S)^{-1} = \omega I^R - I^R H^{RR} g^R (I^R g^R I^R)^{-1}. \tag{15}$$

This demonstrates that the formalism provides a unified description of both material-material interfaces and material-vacuum boundaries.

To establish a connection with the standard surface Green's function formalism, it is useful to recast Eq. (12) in terms of an effective self-energy

$$\Sigma_{\text{eff}} = IHg(IGI)^{-1}, \tag{16}$$

and the interface Green's function can then be written as $(G_I)^{-1} = \omega I - \Sigma_{\text{eff}}$.

⁴For short, we denote $IGI = G_I$ and $I^M g^M I^M = g_I^M$. Particularly, we frequently use the property

$$g_I^M (g_I^M)^{-1} = (g_I^M)^{-1} g_I^M = I^M,$$

which is equivalent to

$$I^M g_I^M = g_I^M I^M = g_I^M, \quad I^M (g_I^M)^{-1} = (g_I^M)^{-1} I^M = (g_I^M)^{-1}.$$

Hence we conclude the following lemmas:

- $(g_I^M)^{-1} g_I^M I^M = (g_I^M)^{-1} g_I^M = I^M$;
- $I^M g_I^M (g_I^M)^{-1} = g_I^M (g_I^M)^{-1} = I^M$.

2.1 In the tight-binding representation

To apply the operator formalism developed above to tight-binding models, it is convenient to express all quantities in the site (Wannier) basis. For a one-dimensional tight-binding system, the bulk Hamiltonian and its Green's function take the form

$$\hat{h} = \sum_{\langle j,j' \rangle} |j\rangle h_{(j,j')} \langle j'| \quad \text{and} \quad \hat{g} = \sum_{j,j'} |j\rangle g_{(j,j')} \langle j'|, \quad (17)$$

where $\langle j, j' \rangle$ denotes the nearest-neighbour pairs. The matrix elements $h_{(j,j')} = \langle j | \hat{h} | j' \rangle$ represents the hopping amplitude from site j' to site j , and $g_{(j,j')}(\omega) = \langle j | \hat{g}(\omega) | j' \rangle = \langle j | (\omega \mathbb{1} - \hat{h})^{-1} | j' \rangle$ is the Green's function describing propagation from site j' to site j at energy ω .

We now consider two semi-infinite tight-binding systems coupled at the interface. Let site $j = 0$ denote the surface of the left subsystem, and $j = 1$ the surface of the right subsystem. The projection operators are then given by

$$\begin{aligned} P^L &= \sum_{j \leq 0} |j\rangle \langle j|, & P^R &= \sum_{j \geq 1} |j\rangle \langle j|, \\ I^L &= |0\rangle \langle 0|, & I^R &= |1\rangle \langle 1|. \end{aligned} \quad (18)$$

We first evaluate the left-left block of $(G_I)^{-1}$. Starting from the operator expression $a = \omega I^L - I^L H P^L g^L (I^L g^L I^L)^{-1}$, one finds

$$\begin{aligned} a &= \omega |0\rangle \langle 0| - \sum_{n,m,n',m',j \leq 0} |0\rangle \langle 0| n \rangle H_{(n,m)} \langle m | j \rangle \langle j | n' \rangle g_{(n',m')}^L \langle m' | 0 \rangle [g_{(0,0)}^L]^{-1} \langle 0| \\ &= \omega |0\rangle \langle 0| - \sum_{j \leq 0} |0\rangle H_{(0,j)} g_{(j,0)}^L [g_{(0,0)}^L]^{-1} \langle 0|. \end{aligned} \quad (19)$$

For a nearest-neighbour tight-binding model, only $j = 0$ and $j = -1$ contribute, yielding

$$a = |0\rangle (\omega - H_{(0,0)} - H_{(0,-1)} g_{(-1,0)}^L [g_{(0,0)}^L]^{-1}) \langle 0| \quad (20)$$

The left-right block is obtained directly as

$$-I^L H I^R = - \sum_{n,m} |0\rangle \langle 0| n \rangle H_{(n,m)} \langle m | 1 \rangle \langle 1| = -|0\rangle H_{(0,1)} \langle 1|, \quad (21)$$

and an analogous derivation applies to the right-right block. Collecting all components, the inverse interface Green's function in the Wannier basis takes the explicit matrix form

$$(G_I)^{-1} = \begin{pmatrix} \omega \mathbb{1} - H_{(0,0)} - H_{(0,-1)} g_{(-1,0)}^L [g_{(0,0)}^L]^{-1} & -H_{(0,1)} \\ -H_{(1,0)} & \omega \mathbb{1} - H_{(1,1)} - H_{(1,2)} g_{(2,1)}^R [g_{(1,1)}^R]^{-1} \end{pmatrix} \quad (22)$$

This representation is particularly convenient for numerical verification: once the bulk properties of the two constituent subsystems and their mutual tunnel coupling are specified, the interface physics can be determined without explicitly solving the full coupled system. Based on this observation, a general procedure for evaluating the interface Green's function of two tight-binding systems can be summarised as follows:

- First, specify the bulk Hamiltonian of each subsystem, either in k-space as h_k^M or in real-space as $h_{(j,j')}^M$ with the two representations related by Fourier transformation.

- Next, compute the bulk Green's function $g_k^M = (\omega \mathbb{1} - h_k^M)^{-1}$, and transform it into real-space to obtain $g_{(j,j')}^M$.
- Then, define the tunnel coupling between the two subsystems through the hopping matrix $V = H_{(1,0)}$, which determines the coupled Hamiltonian $H_{(j,j')}$.
- Finally, substitute these quantities into Eq. (22) to obtain $(G_I)^{-1}$.

Then the local density of states (LDOS) at the interface can be directly calculated as

$$\rho_I(\omega) = -\frac{1}{\pi} \text{Im Tr } G_I(\omega + i0^+). \quad (23)$$

Numerically, the infinitesimal 0^+ is replaced by a finite broadening $\eta = 0.01$, so that the Green's functions are evaluated at $\omega + i\eta$. Isolated poles therefore appear in the plotted LDOS with a small finite linewidth set by this numerical broadening. Matrix inverses are evaluated directly when well conditioned; near singular points, the Moore-Penrose pseudoinverse is used, consistent with Appendix A.3.

Furthermore, by replacing the full trace with a partial trace over the left or right subspace, one can obtain the respective contribution of the left and right subsystems to the total LDOS, and their sum recovers $\rho_I(\omega)$.

Within this framework, the spectral features at the interface are fully determined by the analytical structure of G_I : the isolated poles correspond to localised interface states, appearing as sharp peaks in the LDOS; while the continuous spectrum gives rise to the bulk bands, appearing as smooth, extended region in the LDOS.

Therefore, the evolution of interface states with varying model parameters can be generally understood as the evolution of the pole structure of G_I in the complex energy plane. In the following sections, this perspective will be adopted to analyse various models and tunnel coupling schemes.

3 Application: Su-Schrieffer-Heeger chain

As the simplest one-dimensional tight-binding model exhibiting a topological phase transition, we take the Su-Schrieffer-Heeger (SSH) chain as the first numerical example. Before addressing the coupled problem, we briefly review its essential properties and specify the criteria used in the subsequent analysis.

The SSH model describes a dimerised tight-binding chain with alternating hopping amplitudes. Depending on the relative strength of the intra-cell and inter-cell hoppings, the system resides either in a topologically trivial phase or a non-trivial phase, which can be characterised by the winding number from the bulk Hamiltonian.

Due to the chiral symmetry (after an appropriate energy shift), the energy spectrum of the SSH model is symmetric about the a centre energy $E_c = (\varepsilon_A + \varepsilon_B)/2$. The centre-energy point is therefore of particular significance: its existence is directly connected to the topological property of the system. At an isolated surface, the topologically non-trivial phase supports a surface mode at $E = E_c$ protected by chiral symmetry, whereas the trivial phase does not. [10]

Motivated by this, our analysis focuses on the pole structure of G_I around zero energy and its evolution as system parameters are tuned. In particular, we construct two-dimensional colour

map of the LDOS as a function of the selected model parameter (horizontal axis) and energy (vertical axis), which gives a direct visualisation on the behaviour of interface states.

3.1 Interface Green's function for coupled SSH chain

We begin by specifying the bulk Hamiltonian of a single chain in both real and momentum space. According to Figure 3, the SSH Hamiltonian in real space can be written as

$$\hat{h} = \sum_j \left[\varepsilon_A \hat{c}_{j,A}^\dagger \hat{c}_{j,A} + \varepsilon_B \hat{c}_{j,B}^\dagger \hat{c}_{j,B} - v \left(\hat{c}_{j,A}^\dagger \hat{c}_{j,B} + \text{h.c.} \right) - w \left(\hat{c}_{j+1,A}^\dagger \hat{c}_{j,B} + \text{h.c.} \right) \right], \quad (24)$$

where ε_A and ε_B are the energy offsets of A and B site⁵, v and w denote the intra-cell and inter-cell hopping amplitudes respectively.

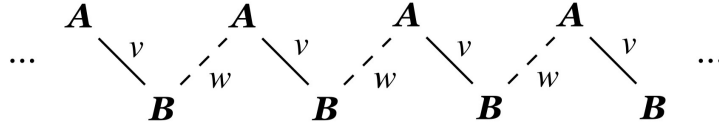


Figure 3: A segment of an infinite SSH chain. A unit cell consists of two sites, A and B, with the hopping amplitudes v, w indicated.

In matrix form, this corresponds to

$$h_{(j,j)} = \begin{pmatrix} \varepsilon_A & -v \\ -v & \varepsilon_B \end{pmatrix}, \quad h_{(j+1,j)} = \begin{pmatrix} 0 & -w \\ 0 & 0 \end{pmatrix}, \quad h_{(j,j+1)} = \begin{pmatrix} 0 & 0 \\ -w & 0 \end{pmatrix}. \quad (25)$$

By using the lattice translational symmetry, we perform a Fourier transform to obtain the Bloch Hamiltonian

$$h_k = \begin{pmatrix} \varepsilon_A & -v - we^{-ika} \\ -v - we^{ika} & \varepsilon_B \end{pmatrix}, \quad (26)$$

from which the bulk Green's function in k -space follows as

$$g_k(\omega) = [\omega \mathbb{1}_2 - h_k]^{-1}. \quad (27)$$

The real-space Green's function is then obtained via inverse Fourier transform,

$$g_{(j,j')}(\omega) = \frac{1}{N} \sum_{k \in \text{BZ}} e^{ik(j-j')a} g_k(\omega), \quad (28)$$

⁵Usually, one sets $\varepsilon_A = \varepsilon_B = \varepsilon$ to simplify the model while preserving spatial inversion symmetry. However, regardless of whether they are equal, any finite on-site energy breaks the chiral symmetry of the system. To keep the model general, we treat ε_A and ε_B as independent parameters here.

which in the continuous limit⁶ $N \rightarrow \infty$ becomes

$$g_{(j,j')}(\omega) = \int_{-\pi}^{\pi} \frac{dk}{2\pi} e^{ik(j-j')a} g_k(\omega). \quad (29)$$

In numerical calculations, the Brillouin-zone integral is evaluated as a discrete Fourier sum on a uniform grid of $N = 2000$ momentum points over $[-\pi, \pi)$. This discretisation follows directly from the finite- N momentum sum in Eq. (28), and is convenient because only the local and nearest-neighbour Green's function element (i.e., $g_{j-j'}$ with $j - j' = 0, \pm 1$) are required in Eq. (22).

Having established the bulk Green's function, we now construct the coupled system. We consider two semi-infinite SSH chains with parameters (v^L, w^L) and (v^R, w^R) , coupled through an interface hopping u , as shown in Figure 4. The full Hamiltonian in real space reads

$$\begin{aligned} \hat{H} = & \sum_{j \geq 1} \left[\varepsilon_A \hat{c}_{j,A}^\dagger \hat{c}_{j,A} + \varepsilon_B \hat{c}_{j,B}^\dagger \hat{c}_{j,B} - v^R \left(\hat{c}_{j,A}^\dagger \hat{c}_{j,B} + \text{h.c.} \right) - w^R \left(\hat{c}_{j+1,A}^\dagger \hat{c}_{j,B} + \text{h.c.} \right) \right] \\ & \sum_{j \leq 0} \left[\varepsilon_C \hat{c}_{j,C}^\dagger \hat{c}_{j,C} + \varepsilon_D \hat{c}_{j,D}^\dagger \hat{c}_{j,D} - v^L \left(\hat{c}_{j,C}^\dagger \hat{c}_{j,D} + \text{h.c.} \right) - w^L \left(\hat{c}_{j+1,C}^\dagger \hat{c}_{j,D} + \text{h.c.} \right) \right] \\ & - u \left(\hat{c}_{1,A}^\dagger \hat{c}_{0,D} + \text{h.c.} \right). \end{aligned} \quad (30)$$

This Hamiltonian can be cast into the same matrix form as the bulk case,

$$H_{(j,j)} = \begin{pmatrix} \varepsilon_\alpha & -v \\ -v & \varepsilon_\beta \end{pmatrix}, \quad H_{(j+1,j)} = \begin{pmatrix} 0 & -w \\ 0 & 0 \end{pmatrix}, \quad H_{(j,j+1)} = \begin{pmatrix} 0 & 0 \\ -w & 0 \end{pmatrix}, \quad (31)$$

with position-dependent parameters

$$v, \varepsilon_\alpha, \varepsilon_\beta = \begin{cases} v^L, \varepsilon_C, \varepsilon_D & j \leq 0 \\ v^R, \varepsilon_A, \varepsilon_B & j > 0 \end{cases}, \quad w = \begin{cases} w^L, & j < 0 \\ u, & j = 0 \\ w^R, & j > 0 \end{cases}. \quad (32)$$

In this way, both the bulk Green's functions g^L, g^R and the coupling Hamiltonian H are explicitly specified. Substituting these into Eq. (22), we obtain G_I for the coupled SSH chains, whose pole structure determines the interface states analysed in the following sections.

Since the topological phase of the SSH model requires the two sublattices to be equivalent, we henceforth set $\varepsilon_C = \varepsilon_D = \varepsilon^L$ and $\varepsilon_A = \varepsilon_B = \varepsilon^R$. The common value $\varepsilon^L, \varepsilon^R$ only shifts the energy origin and does not affect the topological properties.

⁶Under periodic boundary conditions, the allowed k form a discrete set with spacing

$$\Delta k = \frac{2\pi}{Na},$$

where a is the lattice constant (the separation between adjacent unit cells). In the thermodynamic limit $N \rightarrow \infty$, the spacing becomes infinitesimal, $\Delta k \rightarrow dk$, and the sum over k can be replaced by an integral yields

$$\frac{1}{N} \sum_k \rightarrow \frac{a}{2\pi} \int_{-\pi/a}^{\pi/a} dk.$$

In practice, The lattice constant is set to $a = 1$, so that crystal momenta are dimensionless and the Brillouin zone is $k \in [-\pi, \pi)$.

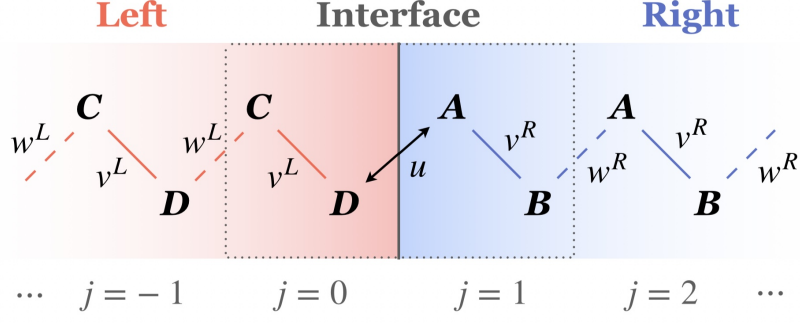


Figure 4: Two semi-infinite SSH chain coupled by tunneling at the interface. The right chain has sites A and B per unit cell, while the left chain has sites C and D. The hopping amplitudes within and between the chains are indicated.

3.2 Evolution of interface physics with tunnel coupling strength

We first consider the case without energy offset (i.e., $\varepsilon^L = \varepsilon^R = 0$) and investigate the evolution of interface states as the tunnel coupling parameter u increases. To illustrate the general mechanism, we select 3 representative configurations:

- Both the left and right sides are in the trivial phase, e.g.,

$$(v^L, w^L) = (0.7, 0.3) \quad \text{and} \quad (v^R, w^R) = (1.2, 0.6).$$

- The left side is in the trivial phase, while the right side is in the topological phase, e.g.,

$$(v^L, w^L) = (0.7, 0.3) \quad \text{and} \quad (v^R, w^R) = (0.6, 1.2).$$

- Both the left and right sides are in the topological phase, e.g.,

$$(v^L, w^L) = (0.3, 0.7) \quad \text{and} \quad (v^R, w^R) = (0.6, 1.2).$$

Note that all energies and hopping amplitudes are expressed in units of a common reference energy scale $E_0 = 1$.

By taking the partial trace over the left and right subspaces separately, one obtain the respective partial LDOS contributions, which sum of which yields the total LDOS. An effectively visualisation method is to encode the LDOS from the left-side using a white-red gradient, and that from the right-side using a white-blue gradient.

For the three configurations described above, the corresponding LDOS can be plotted, as shown in Figure 5. A careful examination of these subplots reveals several key features.

In the weak-coupling limit $u \rightarrow 0$, the interface Green's function reduces to a sum of the surface Green's functions of two independent semi-infinite systems. Consequently, its pole structure is inherited directly from the individual subsystems. This limit provides a well-defined reference point for the subsequent evolution.

As the tunnel coupling parameter increases to a scale comparable to the bulk band of either subsystem, the interface begins to participate in the effective structure of the system. A key observation is that the difference between these three configurations are primarily reflected in the stability of in-gap localised states, particularly those near zero energy.

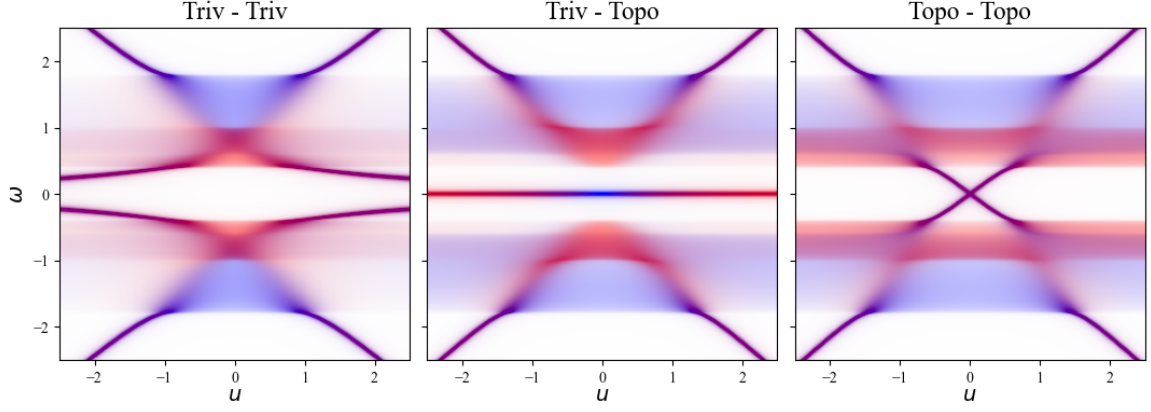


Figure 5: LDOS colour map in the (u, ω) plane for three different configurations. Left: trivial-trivial. Middle: trivial-topological. Right: topological-topological. Brightness encodes the logarithmically scaled LDOS intensity, while hue encodes its left-right distribution: red indicates weight predominantly on the left subsystem and blue indicates weight predominantly on the right subsystem.

To obtain a unified understanding, it is instructive to consider the strong-coupling limit $u \rightarrow \infty$. In this regime, the sites across the interface form a tightly bound dimer due to the large hopping amplitude. This structure is common to all three configurations, and its energy is dominated by u , giving rise to linear high-energy branches in the LDOS that are well separated from the bulk bands. However, the more relevant features lie within the bulk gap. In this limit, the dimer at interface can be regarded as effectively removed from the system, reducing the problem to two truncated SSH chains. The existence of interface states is then determined by the topological properties of the resulting subsystems:

- For the trivial-trivial configuration, the strong bond originally located at the interface is eliminated, leaving both chains effectively terminated by weak bonds. This induces a pair of near-zero-energy states localised near the interface.
- For the trivial-topological configuration, removing one site from each side still results in a topological-trivial junction. As a consequence, the zero-energy mode remains robust throughout the entire range of tunnel coupling strengths.
- For the topological-topological configuration, the zero modes from both sides hybridise under strong tunnel coupling and are pushed away from zero energy. In the end, no in-gap interface states remain.

This analysis shows that, in the strong-coupling regime, the effective topological structure near the interface is redefined.

The qualitative picture developed above can be quantitatively verified by diagonalising minimal interface clusters. We first consider the trivial-trivial case, for which an effective four-site model is constructed from the degrees of freedom near the interface. The corresponding Hamiltonian reads

$$H_{\text{eff}} = - \begin{pmatrix} 0 & v^L & 0 & 0 \\ v^L & 0 & u & 0 \\ 0 & u & 0 & v^R \\ 0 & 0 & v^R & 0 \end{pmatrix}, \quad (33)$$

and the eigenvalues of this matrix are

$$\omega(u) = \pm \sqrt{\frac{(v^L)^2 + (v^R)^2 + (u)^2 \pm \sqrt{[(v^L)^2 + (v^R)^2 + (u)^2]^2 - 4(v^L)^2(v^R)^2}}{2}}. \quad (34)$$

It can be verified that one pair of eigenvalues continuously approaches zero energy as u increases. This branch corresponds to the in-gap interface states observed in the LDOS, which detach from the bulk continuum and evolve towards zero energy in the strong-coupling regime.

For the trivial-topological case, an effective three-site model is sufficient:

$$H_{\text{eff}} = - \begin{pmatrix} 0 & v^L & 0 \\ v^L & 0 & u \\ 0 & u & 0 \end{pmatrix}, \quad (35)$$

and its spectrum is

$$\omega(u) = \pm \sqrt{(v^L)^2 + (u)^2} \quad \text{and} \quad \omega = 0. \quad (36)$$

This result indicates that the zero-energy states remains exactly pinned, consistent with the robustness of the boundary mode.

In the topological-topological case, the minimal description reduces to a two-site model at the interface:

$$H_{\text{eff}} = - \begin{pmatrix} 0 & u \\ u & 0 \end{pmatrix}. \quad (37)$$

The corresponding eigenvalues are

$$\omega(u) = \pm u, \quad (38)$$

implying a complete removal of zero-energy states.

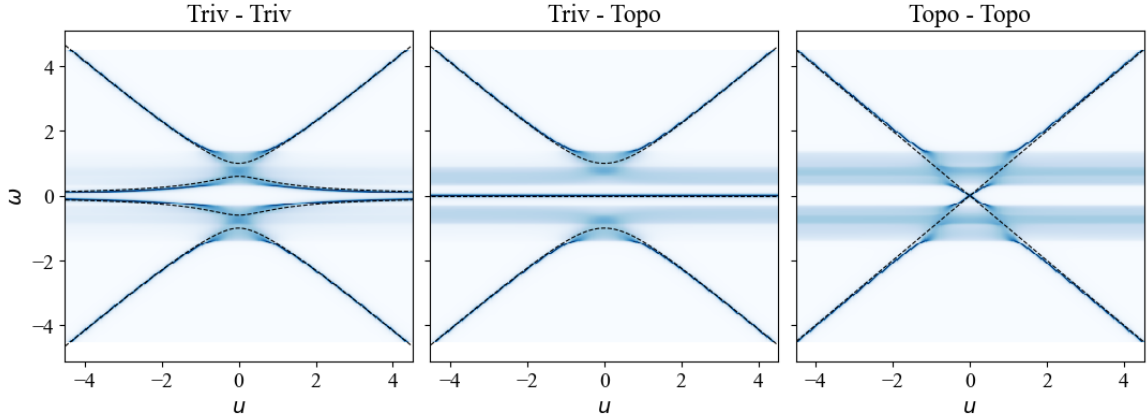


Figure 6: Comparison between the LDOS and the analytical spectra of minimal interface models for the three configurations. The grey dashed curves correspond to Eqs. (33), (34), and (35), respectively. Left: trivial-trivial. Middle: trivial-topological. Right: topological-topological.

By superposing these analytical energy levels onto the LDOS plots, see Figure 6, one finds that they accurately trace the evolution of the interface spectral features away from the bulk band. This demonstrates that, despite the infinite-dimensional nature of the full system, the key behaviour of interface states is largely governed by a finite-dimensional local structure, thereby validating the strong-coupling picture developed above.

3.3 Redistribution of interface state under strong tunnel coupling

In the LDOS plots presented above, contributions from the left and right subsystems are marked in red and blue respectively. For the trivial-topological configuration, a non-trivial phenomenon can be observed: although the zero-energy interface state persists across all tunnel coupling strengths, its spatial distribution undergoes a significant redistribution. In the weak-coupling limit, the state is dominantly localised on the topological side; in contrast, in the strong-coupling limit, the weight almost shifts to the trivial side.

To quantify this behaviour, we introduce a quantity indicating the relative asymmetry of the interface spectral weight between the two sides:

$$\Delta\rho(\omega) = \frac{\rho^L(\omega) - \rho^R(\omega)}{\rho^L(\omega) + \rho^R(\omega)} \quad (39)$$

This quantity takes values in the range $[-1, 1]$, corresponding to states entirely localised on the right or left subsystem. We focus on the zero-energy mode and examine the dependence of $\Delta\rho(0)$ on the tunnel coupling strength u .

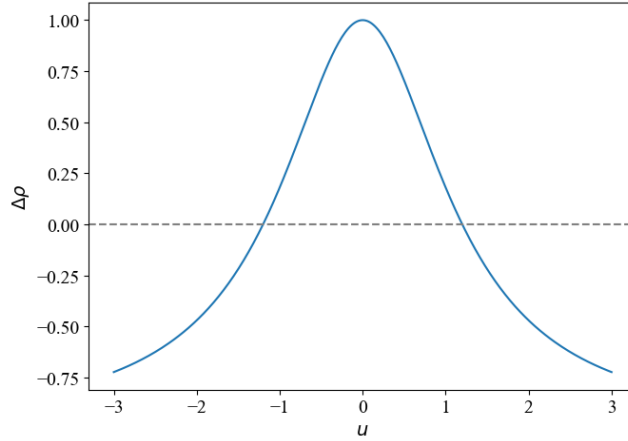


Figure 7: Interface LDOS asymmetry $\Delta\rho(0)$ as a function of the tunnel coupling constant u . The model parameter used is $(v, w) = (0.4, 0.8)$ for the left side (topological phase) and $(v, w) = (1.2, 0.7)$ for the right side (trivial phase).

For the case where the left subsystem is topological and the right subsystem is trivial, the numerical results are shown in Figure 7. One observes that $\Delta\rho(0)$ exhibits a smooth, Lorentzian-like crossover as a function of u . This behaviour can be understood in terms of the mechanism introduced in the last section:

- In the weak-coupling limit, the interfacial state reduces to the isolated surface mode of the topological system, with its wavefunction almost entirely localised on the left side. Accordingly, $\Delta\rho(0) \approx 1$.
- As the tunnel coupling strength increases, the interface state begins to hybridise with the continuum of the trivial subsystem via the hopping term, and gradually leaking into the opposite side. When the tunnel coupling reaches a characteristic scale (numerically comparable to the inter-cell hopping on the trivial side), the contributions from both subsystems become balanced, yielding $\Delta\rho(0) \approx 0$. At this point, the state no longer exhibits obvious localisation on either side.
- In the strong-coupling limit, the hopping across the interface effectively forms a tightly bound dimer, thereby redefining the boundary of the system, as discussed in the previous

section. As a consequence, the spatial position of the zero-energy mode is mainly shifted to the trivial side, causing $\Delta\rho(0) \approx -1$.

This demonstrates that, in a trivial-topological junction, topological protection constrains only the energy of the interface state, while its spatial distribution is governed by the tunnel coupling strength. As the tunnel coupling increases, the interface state undergoes a continuous evolution, from a mode localised mainly on the side, to a delocalised state extending across the interface, and eventually to a state dominated by the opposite trivial side.

3.4 Bulk resonance of surface states

In realistic interfaces, the energy levels of the two materials are generally not aligned. As a consequence, a surface states belonging to one subsystem may lie inside the bulk continuum of the other, in which case it may be no longer a strictly bound state. In this section, we focus on this situation and study how such an embedded interface state evolves as the tunnel coupling strength increases.

We consider the trivial-topological junction and introduce an on-site energy offset⁷ to the topological subsystem, shifting its original zero-energy surface state into the bulk continuum of the trivial subsystem, e.g.,

$$(v^L, w^L, \varepsilon^L) = (1.0, 0.5, 0) \quad \text{and} \quad (v^R, w^R, \varepsilon^R) = (0.4, 1.0, -1.0).$$

Then we can plot a LDOS map for the above coupled systems, as shown in Figure 8.

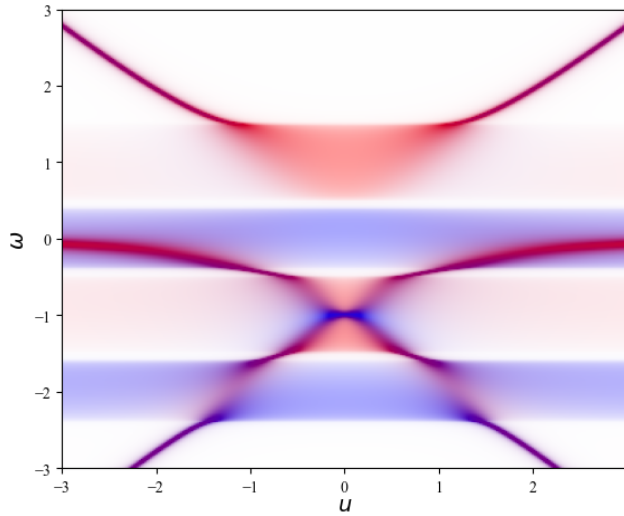


Figure 8: LDOS colour map in the (u, ω) plane for the model above. The model parameter used are $(v^L, w^L, \varepsilon^L) = (1.0, 0.5, 0)$ and $v^R, w^R, \varepsilon^R = (0.4, 1.0, -1.0)$ The colour convention is the same as in Figure 5.

In the weak-coupling limit $u \rightarrow 0$, the interactions between the surface state and the bulk continuum is negligible. Although their energies overlap, the tunnel coupling is too small to induce significant mixing, and the surface state therefore remains visible in the LDOS as a sharp peak corresponding to an isolated localised mode.

⁷Note that introducing an energy offset explicitly breaks the chiral symmetry of the SSH model. Consequently, strictly speaking, the edge states are therefore no longer symmetry-protected. However, for small shifts, the edge states remain well localised.

As the tunnel coupling strength increases to the scale of the bulk bandwidth, the surface state begins to interact strongly with the continuum. The originally sharp peak gradually broadens, which spectral weight is redistributed within the continuum, leading to an enhanced response near the band edges. This process can be interpreted as the leakage of the surface state into the bulk states of the trivial subsystem, forming a resonance embedded in the continuum.

Then consider strong-coupling limit $u \rightarrow \infty$. Numerical results show that, apart from the high-energy dimer states induced by strong tunnel coupling at the interface, a sharp peak re-emerges near zero energy, which is effectively decoupled from the bulk continuum. This zero-energy feature can again be understood from a minimal interface cluster. In the limit $u \rightarrow \infty$, the dominant interface Hamiltonian reduces to

$$H_{\text{eff}} = - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & u \\ 0 & u & 0 \end{pmatrix}. \quad (40)$$

This matrix has three eigenvalues:

$$\omega(u) = \pm u \quad \text{and} \quad \omega = 0. \quad (41)$$

The eigenvalues $\pm u$ correspond to the strongly coupled dimer, while the zero-energy state originates from an unpaired local degree of freedom that remains after strong mixing.

In summary, the interface state undergoes a crossover from a sharp bound state to a broadened resonance in the continuum, and ultimately re-emerges as an isolated zero-energy modes in the strong-coupling regime.

4 Application: $p + ip$ superconductor

A superconducting system can be formally viewed as an effective band structure within the Bogoliubov-de Gennes (BdG) framework. In the superconducting phase, electrons form Cooper pairs due to an effective attractive interaction, opening a pairing gap near the Fermi level. Similar to band insulators, this gap can host non-trivial topological structures, thereby giving rise to topological superconductivity.[11]

The above topological framework can be naturally extended to two-dimensional systems. As a typical model of a two-dimensional topological superconductor, the $p + ip$ superconductor will serve as the central example in this section. In contrast to the SSH model, where zero-energy surface states are protected by chiral symmetry, the topological features of the $p + ip$ model manifest as chiral modes propagating along the interface. Their chirality originates from the momentum dependent complex pairing term $\Delta(\mathbf{k})$ which carries a non-trivial winding structure. Specifically, this complex pairing acquires a non-trivial phase winding as a function of momentum direction, thereby breaking the time-reversal symmetry and inducing a non-zero topological invariant, the Chern number. By the bulk-boundary correspondence, this implies the existence of uni-directional chiral edge states, which are realised as Majorana boundary states in superconducting systems. [12]

Depending on the model parameters, the $p + ip$ system can have different topological phases.

- When the Chern number $C = 0$, the system is topologically trivial. In certain parameter regimes, localised modes may still occur within the gap; however, their dispersion forms closed loops within the Brillouin zone and does not connect the bulk bands, and hence they are not topologically protected.

- When the Chern number $C = \pm 1$, the system is topological non-trivial. In this case, chiral boundary necessarily emerge at the interface, and their dispersion spans the entire bulk gap and connects the lower and upper bands. This appears as a continuous dispersion branch traversing the gap. The sign of its slope, corresponding to the group velocity, determines the propagation direction: a positive slope indicates motion in the positive direction, while a negative slope indicates the opposite.

Accordingly, in the following analysis, the topological nature of interface states is identified by whether the in-gap dispersion connects the bulk bands, while the sign of its slope encodes the direction of propagation.

4.1 Interface Green's function for Coupled $p + ip$ superconductor

We now extend the general framework developed in section 2.1 to the $p + ip$ superconducting system.

Compared to the SSH model, the key difference is the superconducting pairing, which requires a BdG description. In the mean-field approximation, the Hamiltonian can be written in Nambu space as

$$\hat{h} = \sum_{\mathbf{k}} \begin{pmatrix} \hat{c}_{\mathbf{k},\uparrow}^\dagger & \hat{c}_{-\mathbf{k},\downarrow} \end{pmatrix} \begin{pmatrix} \varepsilon_{\mathbf{k}} - \mu & \Delta_{\mathbf{k}} \\ \Delta_{\mathbf{k}}^* & -(\varepsilon_{-\mathbf{k}} - \mu) \end{pmatrix} \begin{pmatrix} \hat{c}_{\mathbf{k},\uparrow} \\ \hat{c}_{-\mathbf{k},\downarrow}^\dagger \end{pmatrix}, \quad (42)$$

which defines an effective two-band Hamiltonian in particle-hole space. This allows the superconducting system to be treated on the same way as an insulating band structure.

For the chiral p -wave superconductor [13], we take

$$\begin{aligned} \varepsilon_{\mathbf{k}} &= \varepsilon_0 - 2t_x \cos k_x - 2t_y \cos k_y, \\ \Delta_{\mathbf{k}} &= \Delta(\sin k_x - i\alpha \sin k_y), \end{aligned} \quad (43)$$

where $\alpha = \pm 1$ corresponds to the two chiralities. The Hamiltonian can be written in the compact form

$$h_{\mathbf{k}} = \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad (44)$$

with

$$\mathbf{d}(\mathbf{k}) = \begin{pmatrix} \Delta \sin k_x \\ \alpha \Delta \sin k_y \\ \varepsilon_0 - \mu - 2t_x \cos k_x - 2t_y \cos k_y \end{pmatrix}. \quad (45)$$

The bulk Green's function then follows as

$$g_{\mathbf{k}}(\omega) = [\omega \mathbb{1} - \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}] = \frac{\omega \mathbb{1} + \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}}{\omega^2 - |\mathbf{d}(\mathbf{k})|^2}. \quad (46)$$

To construct a semi-infinite system, we perform a partial Fourier transform along the y -direction, keeping k_x as a good quantum number. This maps the original two-dimensional system to a set of effectively one-dimensional chains labeled by k_x . The real-space Green's function along the direction normal to the interface is then given by

$$g_{k_x, (j, j')}(\omega) = \frac{1}{N} \sum_{k_y} e^{ik_y(j-j')a} g_{\mathbf{k}}(\omega), \quad (47)$$

which plays the role of the bulk Green's function in the SSH analysis. Numerically, it will use the same discretisation as in Section 3.

With this construction, the problem reduces to coupling two semi-infinite k_x -resolved chains. By taking the partial Fourier transform, the Hamiltonian can be cast into

$$\begin{aligned} h_{k_x, (j,j)} &= \begin{pmatrix} \varepsilon_0 - \mu - 2t_x \cos k_x & \Delta \sin k_x \\ \Delta \sin k_x & -\varepsilon_0 + \mu + 2t_x \cos k_x \end{pmatrix}, \\ h_{k_x, (j+1,j)} &= \begin{pmatrix} -t_y & \alpha\Delta/2 \\ -\alpha\Delta/2 & t_y \end{pmatrix}, \\ h_{k_x, (j,j+1)} &= \begin{pmatrix} -t_y & -\alpha\Delta/2 \\ \alpha\Delta/2 & t_y \end{pmatrix}. \end{aligned} \quad (48)$$

Hereafter, the superscripts L (for $j \leq 0$) and R (for $j \geq 1$) are adopted to label the model parameters of the left and right semi-infinite systems respectively, to distinguish their respective degrees of freedom.

Analogous to the inter-layer hopping matrices in the bulk Hamiltonian, we construct the interface hopping matrices in the same structural form:

$$H_{k_x, (1,0)} = \begin{pmatrix} -t_c & \Delta_c/2 \\ -\Delta_c/2 & t_c \end{pmatrix}, \quad H_{k_x, (0,1)} = \begin{pmatrix} -t_c & -\Delta_c/2 \\ \Delta_c/2 & t_c \end{pmatrix}, \quad (49)$$

where t_c denotes the conventional normal hopping amplitude and Δ_c represents the pairing-induced superconducting coupling across the interface between the left and right subsystems. In the following, we focus on the case $\Delta_c = 0$, where only normal hopping is present across the interface.

At this point, all ingredients required by the general recipe are specified: the bulk Green's functions $g_{k_x}^L, g_{k_x}^R$ and the real-space coupled Hamiltonian. Substituting these into the expression for the interface Green's function $G_I(k_x, \omega)$, we obtain the full momentum-resolved interface Green's function, whose pole structure determines the dispersive interface states analysed in the subsequent sections.

4.2 Hybridisation between coupled edge states

Now, we analyse the interface spectra for different topological configurations of two coupled $p + ip$ superconductors.

Unlike the SSH case, here the relevant objects are no longer an isolated zero poles, but dispersive chiral modes $\omega(k_x)$. To make contact with the previous analysis, this is useful to regard k_x as a parameter and consider, for each fixed k_x , the pole structure of G_I as a function of ω . In this way, the interface spectrum is obtained by assembling these poles into continuous branches in (k_x, ω) space.

We now consider several representative coupling configurations of $p + ip$ superconductors, and compute the LDOS at interface during the same numerical procedure as described previously.

We begin with the case where the two subsystems have identical Chern numbers. Taking $C^L = C^R = 1$ as an example, each subsystem supports a chiral edge mode, but with opposite propagation directions. The corresponding LDOS at three representative tunnel coupling strengths is shown in Figure 9 where we show also the chosen bulk parameters for the left and right subsystems.

In the weak-coupling limit, the edge modes on the two sides are effectively decoupled. Their dispersions intersect at $k_x = 0$, forming a zero-energy degeneracy. As the tunnel coupling is

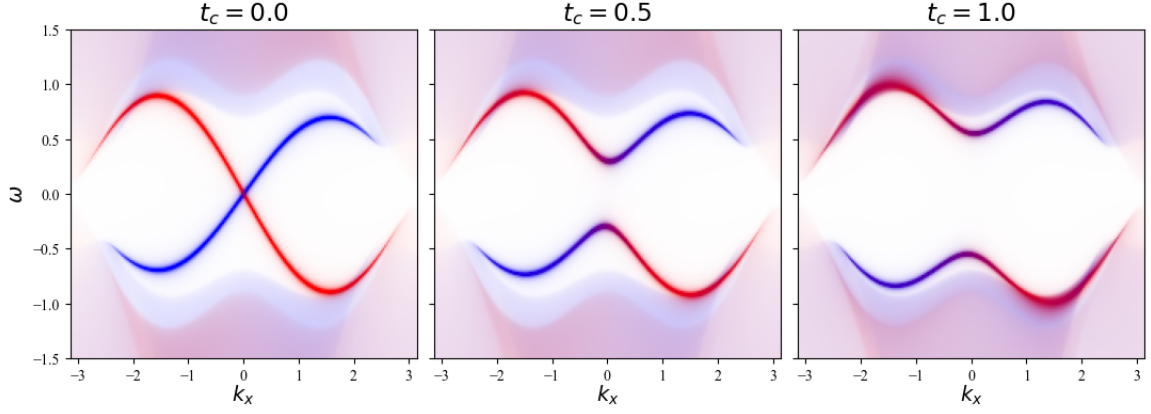


Figure 9: LDOS colour map in the (k_x, ω) plane for different tunnel coupling parameters t_c . The model parameters used are $(\mu, \Delta, t_x, t_y) = (-1.5, 0.9, 1.0, 1.7)$ for the left subsystem (red), and $(\mu, \Delta, t_x, t_y) = (-1.7, 0.7, 1.0, 1.6)$ for the right subsystem (blue). The colour convention is the same as in Figure 5.

turned on, these two modes hybridise near the degeneracy point, forming an effective two-level system. As a result, the crossing is lifted and a gap opens, leading to a fully gapped interface spectrum. The interface spectrum therefore becomes fully gapped.

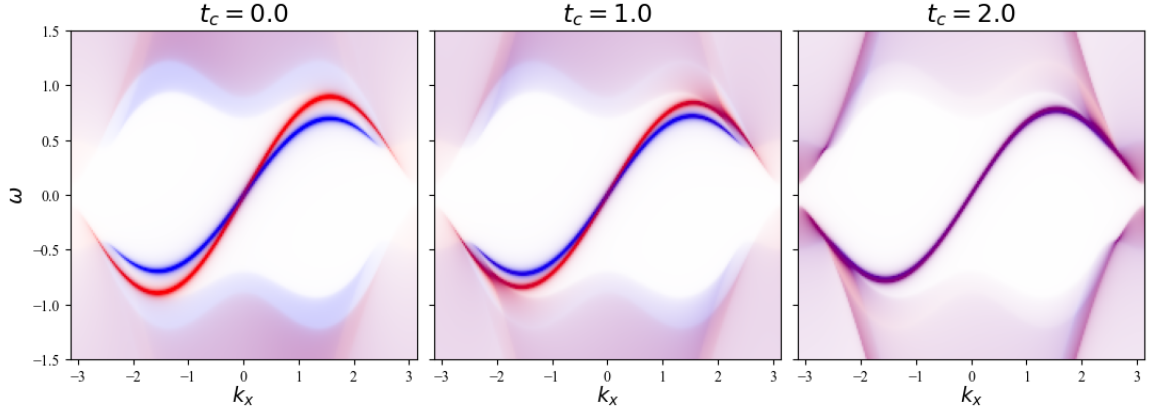


Figure 10: LDOS colour map in the (k_x, ω) plane for different tunnel coupling parameters t_c . The model parameters used are $(\mu, \Delta, t_x, t_y) = (-1.5, 0.9, 1.0, -1.7)$ for the left subsystem (red), and $(\mu, \Delta, t_x, t_y) = (-1.7, 0.7, 1.0, 1.6)$ for the right subsystem (blue). The colour convention is the same as in Figure 5.

Next, consider the case of opposite Chern numbers, e.g., $C^L = -1$ and $C^R = 1$. In this configuration, the chiral edge modes of the two subsystems propagate in the same direction. The corresponding LDOS, shown in Figure 10, indicates that no gap opens near $k_x = 0$, even for a much stronger tunnel coupling. As t_c increases, the two dispersive branches approach each other but remain gapless throughout.

This contrast demonstrates that, the behaviour of chiral edge modes under tunnel coupling is governed by their relative propagation directions. To test this mechanism in a more general setting, we further consider the tunnel coupling between a chiral edge mode and an in-gap localised state in a topologically trivial material.

When the trivial side hosts an in-gap state who propagate in the opposite direction to the chiral edge state, Figure 11 shows that a gap-opening behaviour similar to the previous case occurs when tunnel coupling opens. However, since the in-gap state forms a closed dispersion within

the Brillouin zone, the resulting interface spectrum still connects the lower and upper bands (even though it is across the full Brillouin zone), and a gap-spanning interface branch remains present. This is consistent with the persistence of a topological interface mode.

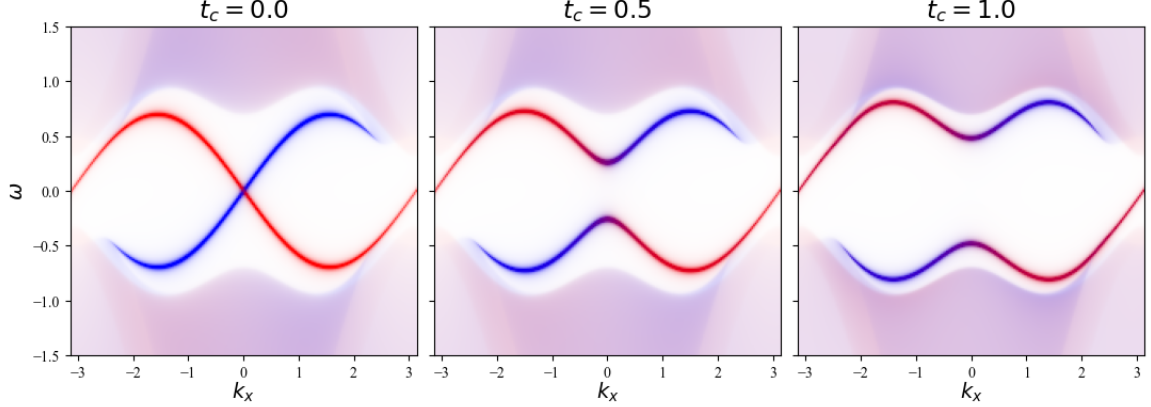


Figure 11: LDOS colour map in the (k_x, ω) plane for different tunnel coupling parameters t_c . The model parameters used are $(\mu, \Delta, t_x, t_y) = (-1.7, 0.7, 1.0, 2.1)$ for the left subsystem (red), and $(\mu, \Delta, t_x, t_y) = (-1.7, 0.7, 1.0, 1.6)$ for the right subsystem (blue). The colour convention is the same as in Figure 5.

On the other hand, If the in-gap state propagate in the same direction as the chiral edge mode, no significant hybridisation is observed even at strong tunnel coupling, as shown in Figure 12.

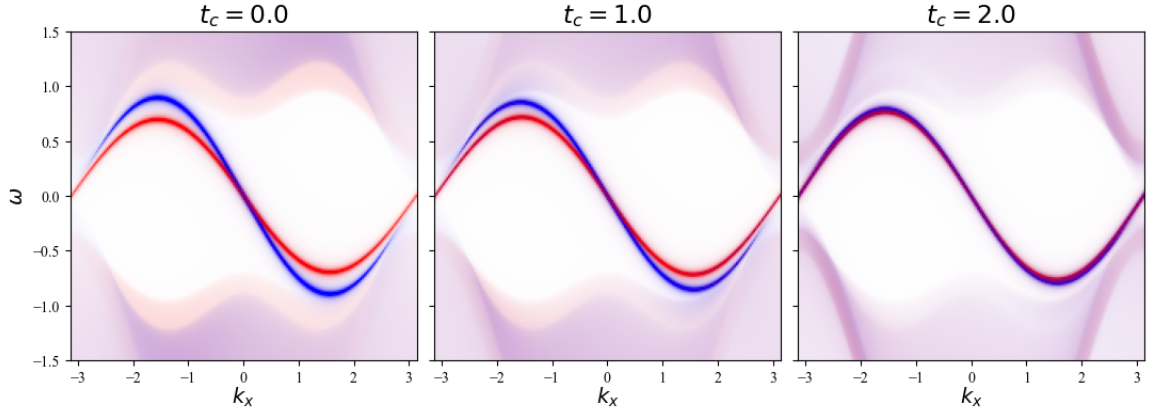


Figure 12: LDOS colour map in the (k_x, ω) plane for different tunnel coupling parameters t_c . The model parameters used are $(\mu, \Delta, t_x, t_y) = (-1.7, 0.7, 1.0, 2.1)$ for the left subsystem (red), and $(\mu, \Delta, t_x, t_y) = (-1.5, 0.9, 1.0, -1.7)$ for the right subsystem (blue). The colour convention is the same as in Figure 5.

To sum up, these results reveals a unified mechanism: whether a gap opens under tunnel coupling is determined by the existence of a pair of low-energy modes that meets in k -space with opposite propagation directions. When this condition is satisfied, the tunnel coupling induces effective hybridisation and opens a gap; otherwise, the tunnel coupling leads only to a deformation of the dispersion, without destroying the gapless structure.

5 Conclusion

In this work, we have reformulated and applied the interface Green's function formalism of Garcia-Moliner and Velasco for coupled tight-binding systems, and use it to analyse interface

states across different tunnel-coupling regimes. By expressing the interface Green's function entirely in terms of bulk quantities, the interfacial problem is reduced to a projection of the uncoupled subsystems, avoiding the need for explicit diagonalisation of the full system.

With this formulation, we can produce a unified physical picture for the evolution of interface states. Across different models, the effect of tunnel coupling can be understood as a competition between bulk structure and interface physics: in the weak-coupling regime, interface states inherit their properties from the individual subsystem, while in the strong-coupling regime, the interface undergoes an effective reconstruction governed by local cluster configuration. Another key outcome is the qualitative changes in the interface spectrum. In particular, gap opening is associated with the presence of counter-propagating modes at the same momentum, while robust gapless behaviour persists when such mixing channels are absent.

From a wider perspective, the interface does not introduce fundamentally new degrees of freedom, but rather reorganises existing ones in a highly constrained manner. This view provides a compact and physically transparent framework for analysing coupled quantum systems, and may serve as a useful starting point for more complex setting, including higher-dimensional interfaces and interacting systems.

Appendix

A.1 Algebraic Properties of Identity Operators and Projectors

Denote M to be either side of the system and $\bar{M}$ to be the other side (e.g., if M is L , then $\bar{M}$ is R), and take $M' \in \{M, \bar{M}\}$. The projectors satisfy the following fundamental relations:

- **Hermiticity:** The identity operator and projectors are self-adjoint, i.e.,

$$\mathbb{1}^\dagger = \mathbb{1}, \quad (P^M)^\dagger = P^M, \quad I^\dagger = I, \quad (I^M)^\dagger = I^M.$$

- **Idempotency:** The identity operator and projectors are idempotent, i.e.,

$$\mathbb{1}^2 = \mathbb{1}, \quad (P^M)^2 = P^M, \quad I^2 = I, \quad (I^M)^2 = I^M.$$

- **Inclusion:** Projectors associated with the same subspace exhibit a nested structure, i.e.,

$$P^M \mathbb{1} = \mathbb{1} P^M = P^M, \quad I^M P^M = P^M I^M = I^M.$$

- **Orthogonality:** Projectors associated with different subspaces are mutually orthogonal, i.e.,

$$P^M P^{\bar{M}} = 0, \quad I^M I^{\bar{M}} = 0, \quad I^M P^{\bar{M}} = P^{\bar{M}} I^M = 0.$$

We can write the last 3 properties into a more compact form: We can write the last 3 properties into a more compact form

$$\begin{aligned} P^M P^{M'} &= \delta^{MM'} P^M, \\ I^M I^{M'} &= \delta^{MM'} I^M, \\ I^M P^{M'} &= P^{M'} I^M = \delta^{MM'} I^M, \end{aligned}$$

where $\delta^{MM'}$ are Kronecker delta, i.e.,

$$\delta^{MM'} = \begin{cases} 1, & M' = M; \\ 0, & M' = \bar{M}. \end{cases}$$

Hence, we can deduce a useful lemma:

$$P^M I = I P^M = I^M, \quad (50)$$

since

$$\begin{aligned} P^M I &= P^M (I^M + I^{\bar{M}}) = \underbrace{P^M I^M}_{=I^M} + \underbrace{P^M I^{\bar{M}}}_{=0} = I^M, \\ I P^M &= (I^M + I^{\bar{M}}) P^M = \underbrace{I^M P^M}_{=I^M} + \underbrace{I^{\bar{M}} P^M}_{=0} = I^M. \end{aligned}$$

A.2 Block Structure of Operators

Since the full Hilbert space admits $\mathcal{H} = \mathcal{H}^L \oplus \mathcal{H}^R$, any linear operator $A \in \mathcal{L}(\mathcal{H})$ can be decomposed as a sum of its projections:

$$A = \sum_{M, M'} P^M A P^{M'} = P^L A P^L + P^L A P^R + P^R A P^L + P^R A P^R.$$

For convenience, A can possess a natural block structure:

$$A = \begin{pmatrix} A^{LL} & A^{LR} \\ A^{RL} & A^{RR} \end{pmatrix} \equiv \begin{pmatrix} P^L A P^L & P^L A P^R \\ P^R A P^L & P^R A P^R \end{pmatrix}.$$

where the blocks $A^{MM'} \equiv P^M A P^{M'}$ acts as a linear map from subspace $\mathcal{H}^{M'}$ to $\mathcal{H}^M$, with matrix dimension of $d^M \times d^{M'}$.

If we project the operator onto the interface, it will possess a similar block structure

$$I A I \equiv A_I = \begin{pmatrix} A_I^{LL} & A_I^{LR} \\ A_I^{RL} & A_I^{RR} \end{pmatrix} \equiv \begin{pmatrix} I^L A I^L & I^L A I^R \\ I^R A I^L & I^R A I^R \end{pmatrix}.$$

where the blocks $A_I^{MM'} \equiv I^M A I^{M'} = I A^{MM'} I$ denote the operator A projecting onto the corresponding interface.

Under this convention, the identity operator $\mathbb{1}$ and interface projector I are block-diagonal with respect to the L - R decomposition, i.e.,

$$\begin{aligned} \mathbb{1} &= P^L + P^R = \begin{pmatrix} P^L & 0 \\ 0 & P^R \end{pmatrix}, \\ I &= I^L + I^R = \begin{pmatrix} I^L & 0 \\ 0 & I^R \end{pmatrix}. \end{aligned}$$

A.3 Moore-Penrose Pseudoinverse

If we use the block structure described above, the blocks are singular in the full Hilbert space $\mathcal{H}$. When an "inverse" is required for a singular block N , it is defined as the Moore-Penrose pseudoinverse N^+ on the appropriate support, such that:

- $(NN^+)N = N$;
- $(N^+N)N^+ = N^+$.

Note that N^+N and NN^+ are Hermitian and idempotent operators. [14]

Simply conclude, in our case, if a matrix A is in the form of

$$A = \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} \quad \text{or} \quad A = \begin{pmatrix} B & 0 \\ 0 & C \end{pmatrix},$$

where B and C are invertible within the corresponding subspace $\mathcal{H}^B$ and $\mathcal{H}^C$ respectively, then we can denote its pseudoinverse as simply

$$A^+ = \begin{pmatrix} B^{-1} & 0 \\ 0 & 0 \end{pmatrix} \quad \text{or} \quad A^+ = \begin{pmatrix} B^{-1} & 0 \\ 0 & C^{-1} \end{pmatrix}.$$

In the former case, we have $AA^+ = A^+A = P^B$, where P^B is the projector of the subspace $\mathcal{H}^B$; in the latter case, $AA^+ = A^+A = \mathbb{1}$.

For convenience and convention, we still use the notation $(\dots)^{-1}$.

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